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WINDOWS FUNDAMENTAL ON HACK THE BOX

1.1 Introduction

Windows Fundamentals is a module designed to strengthen my understanding of the Windows operating system by teaching how to perform essential system tasks and manage daily operations in a Windows environment. It focuses on practical skills such as user and group management, configuring system services, and—most importantly—correctly granting and controlling permissions so that the right users and groups have appropriate access to resources. This knowledge is crucial for maintaining system security, enforcing the principle of least privilege, and ensuring proper access control in enterprise environments.

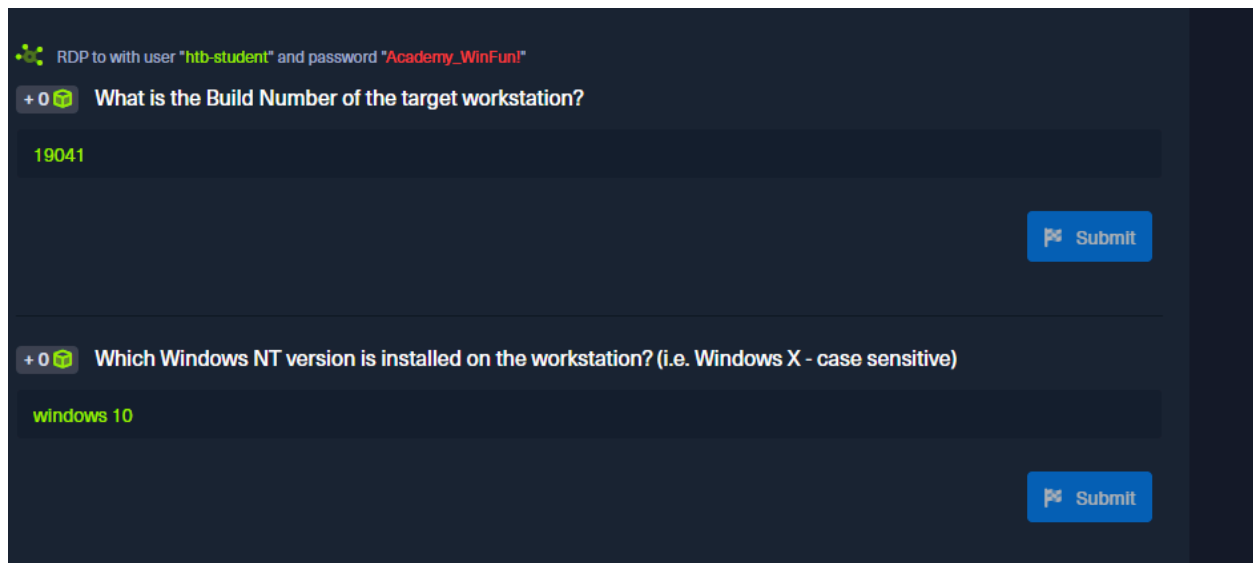
1.2 Introduction to Windows

I learnt about the fundamentals of windows operating systems, when the first windows were invented and the major features that were added to the windows that constitute the latest windows version.

Questions:

1. What is the Build Number of the target workstation?
2. Which Windows NT version is installed on the workstation?

Answers as shown in the diagram below:



The screenshot shows a terminal window with a dark background. At the top, it says "RDP to with user 'htb-student' and password 'Academy_WinFun!'". Below this, there are two questions, each preceded by a green icon with a plus sign and a lock icon. The first question is "What is the Build Number of the target workstation?" and the answer "19041" is entered in the text field. The second question is "Which Windows NT version is installed on the workstation? (i.e. Windows X - case sensitive)" and the answer "windows 10" is entered in the text field. Both questions have a blue "Submit" button to their right.

RDP to with user "htb-student" and password "Academy_WinFun!"

+ 0 What is the Build Number of the target workstation?

19041

Submit

+ 0 Which Windows NT version is installed on the workstation? (i.e. Windows X - case sensitive)

windows 10

Submit

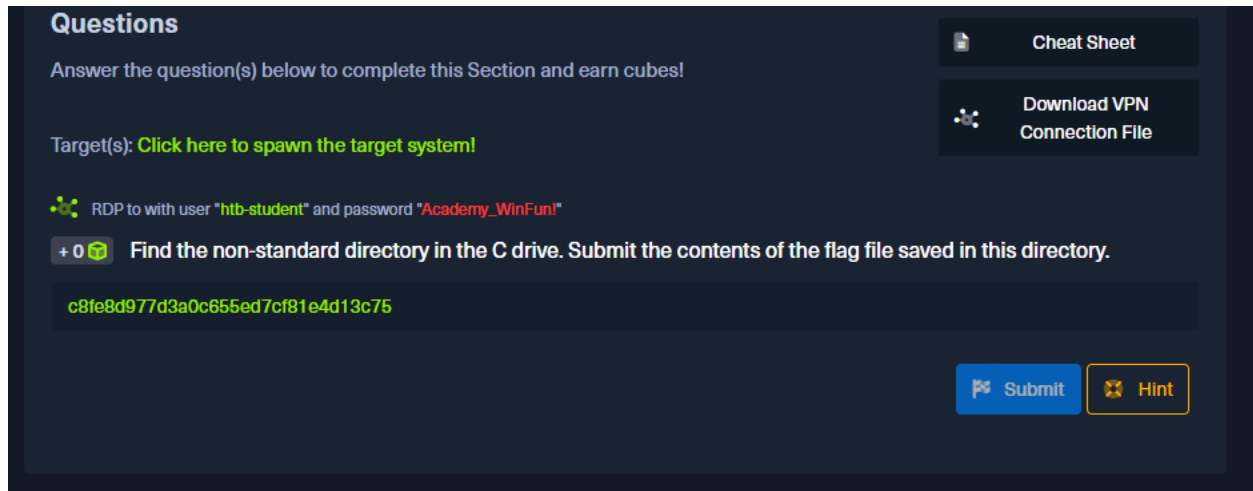
1.3 OPERATING SYSTEMS

Under operating systems , I navigated the c directory which is also the root directory. I explored the dir command.

Question:

1.Find the non-standard directory in the C drive. Submit the contents of the flag file saved in this directory.

Answer as shown below:



The screenshot shows a HackTheBox question interface. At the top, it says "Questions" and "Answer the question(s) below to complete this Section and earn cubes!". On the right, there are two buttons: "Cheat Sheet" and "Download VPN Connection File". The question text is "Find the non-standard directory in the C drive. Submit the contents of the flag file saved in this directory." Above the question, there is a target system link: "Click here to spawn the target system!". Below the question, there is a hint: "RDP to with user 'htb-student' and password 'Acaderry_WinFun!'". At the bottom, there is a "Submit" button and a "Hint" button. The flag value "c8fe8d977d3a0c655ed7cf81e4d13c75" is displayed in a green box.

1.4 FILE SYSTEMS

I learnt of the old file systems the FAT12, FAT16, FAT32, NTFS, and exFAT. FAT12 and FAT16 .The modern file system is called NTFS file system.

Under NTFS, I learnt of the pros and cons of the NTFS file system nad the example of permissions.

NTFS (New Technology File System) is the default Windows file system since Windows NT 3.1. In addition to making up for the shortcomings of FAT32, NTFS also has better support for metadata and better performance due to improved data structuring.

Pros of NTFS:

- NTFS is reliable and can restore the consistency of the file system in the event of a system failure or power loss.
- Provides security by allowing us to set granular permissions on both files and folders.
- Supports very large-sized partitions.
- Has journaling built-in, meaning that file modifications (addition, modification, deletion) are logged.

Cons of NTFS:

- Most mobile devices do not support NTFS natively.
- Older media devices such as TVs and digital cameras do not offer support for NTFS storage devices.

Question:

What system user has full control over the c:\users directory?

Answer

The code i used as shown in the diagram:

```
Windows PowerShell
PS C:\Academy> icacls C:\Users
>>
C:\Users Everyone:(OI)(CI)(RX)
          NT AUTHORITY\SYSTEM:(OI)(CI)(F)
          BUILTIN\Administrators:(OI)(CI)(F)
          WS01\bob.smith:(OI)(CI)(F)
          BUILTIN\Users:(OI)(CI)(RX)

Successfully processed 1 files; Failed processing 0 files
PS C:\Academy>
```

Questions

Answer the question(s) below to complete this Section and earn cubes!

Target(s): [Click here to spawn the target system!](#)

RDP to with user "**htb-student**" and password "***Academy_WinFun!**"

+ 0 What system user has full control over the c:\users directory?

bob.smith

Submit

Cheat Sheet

Download VPN Connection File

1.5 NTFS VS .SHARE PERMISSIONS

In this module i learnt about share permissions in a folder. The Server Message Block protocol (SMB) is used in Windows to connect shared resources like files and printers. It is used in large, medium, and small enterprise environments. See below the visual on sharing

Share permissions

Permission	Description
Full Control	Users are permitted to perform all actions given by Change and Read permissions as well as change permissions for NTFS files and subfolders
Change	Users are permitted to read, edit, delete and add files and subfolders
Read	Users are allowed to view file & subfolder contents

NTFS Basic permissions

Permission	Description
Full Control	Users are permitted to add, edit, move, delete files & folders as well as change NTFS permissions that apply to all allowed folders
Modify	Users are permitted or denied permissions to view and modify files and folders. This includes adding or deleting files
Read & Execute	Users are permitted or denied permissions to read the contents of files and execute programs
List folder contents	Users are permitted or denied permissions to view a listing of files and subfolders
Read	Users are permitted or denied permissions to read the contents of files

See below on special characters:

NTFS special permissions

Permission	Description
Full control	Users are permitted or denied permissions to add, edit, move, delete files & folders as well as change NTFS permissions that apply to all permitted folders
Traverse folder / execute file	Users are permitted or denied permissions to access a subfolder within a directory structure even if the user is denied access to contents at the parent folder level. Users may also be permitted or denied permissions to execute programs
List folder/read data	Users are permitted or denied permissions to view files and folders contained in the parent folder. Users can also be permitted to open and view files
Read attributes	Users are permitted or denied permissions to view basic attributes of a file or folder. Examples of basic attributes: system, archive, read-only, and hidden
Read extended attributes	Users are permitted or denied permissions to view extended attributes of a file or folder. Attributes differ depending on the program
Create files/write data	Users are permitted or denied permissions to create files within a folder and make changes to a file
Create folders/append data	Users are permitted or denied permissions to create subfolders within a folder. Data can be added to files but pre-existing content cannot be overwritten
Write attributes	Users are permitted or denied to change file attributes. This permission does not grant access to creating files or folders

For my assignment this module , i hade 3 questions , which are answered as below:

RDP to 10.129.61.46 (ACADEMY-MISC-WS01) with user "htb-student" and password "Academy_WinFun!"

+ 1 🟢 What protocol discussed in this section is used to share resources on the network using Windows? (Format: case sensitive)

SMB

Submit Hint

+ 1 🟢 What is the name of the utility that can be used to view logs made by a Windows system? (Format: 2 words, 1 space, not case sensitive)

event viewer

Submit Hint

+ 1 🟢 What is the full directory path to the Company Data share we created?

C:\Users\htb-student\Desktop\Company Data

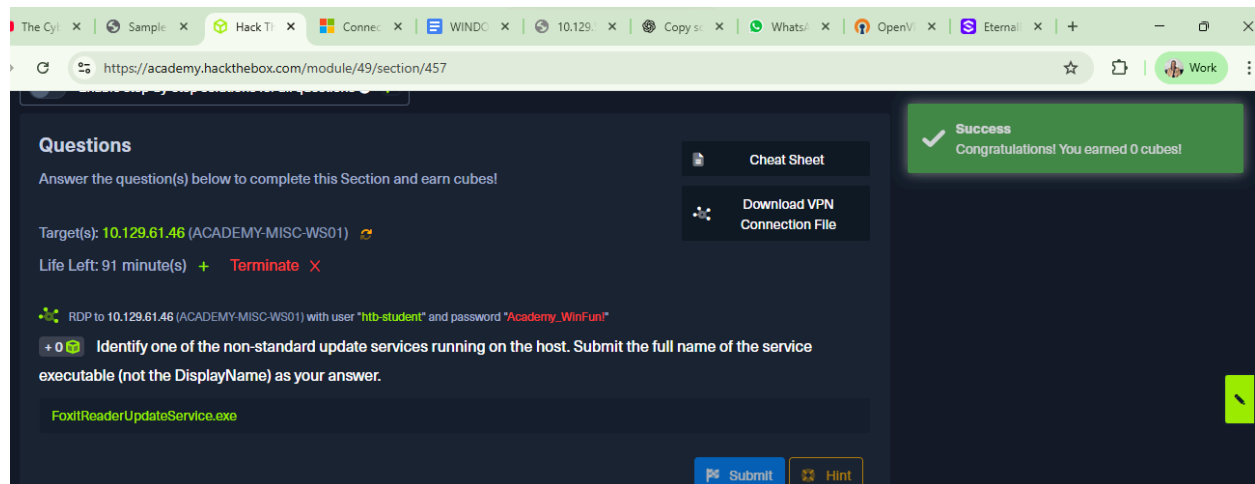
✓ Success
Congratulations! You earned 1 cube!

1.6 WINDOW SERVICES AND PROCESSES

Services on Windows are responsible for many functions within the Windows operating system, such as networking functions, performing system diagnostics, managing user credentials, controlling Windows updates, and more. Windows services are managed via the Service Control Manager (SCM) system, accessible via the services.msc MMC add-in.

Question

Identify one non-standard update running on the host, below is evidence of my answer as illustrated below:



1.7 Window session

I learnt about interactive and non-interactive windows sessions.

An interactive, or local logon session, is initiated by a user authenticating to a local or domain system by entering their credentials. **Eg., through powershell, remote desktop connections, cmd.**

And finally learnt on how to get aliases from the windows powershell , as shown below:

```
Windows PowerShell
>>
PS C:\Users\htb-student> Get-Alias

CommandType      Name
-----
Alias             % -> ForEach-Object
Alias             ? -> Where-Object
Alias             ac -> Add-Content
Alias             asnp -> Add-PSSnapin
Alias             cat -> Get-Content
Alias             cd -> Set-Location
Alias             CFS -> ConvertFrom-String
Alias             chdir -> Set-Location
Alias             clc -> Clear-Content
Alias             clear -> Clear-Host
Alias             clhy -> Clear-History
Alias             cli -> Clear-Item
Alias             clp -> Clear-ItemProperty
Alias             cls -> Clear-Host
Alias             clv -> Clear-Variable
Alias             cnsn -> Connect-PSSession
Alias             compare -> Compare-Object
Alias             copy -> Copy-Item
Alias             cp -> Copy-Item
Alias             cpi -> Copy-Item
Alias             cpp -> Copy-ItemProperty
Alias             curl -> Invoke-WebRequest
Alias             cvpa -> Convert-Path
Alias             dbp -> Disable-PSBreakpoint
Alias             del -> Remove-Item
Alias             diff -> Compare-Object
Alias             dir -> Get-ChildItem
Alias             dsn -> Disconnect-PSSession
Alias             ebp -> Enable-PSBreakpoint
Alias             echo -> Write-Output
Alias             epal -> Export-Alias
Alias             epsv -> Export-Csv
Alias             epsn -> Export-PSSession
Alias             erase -> Remove-Item
Alias             etsn -> Enter-PSSession
Alias             exsn -> Exit-PSSession
Alias             fc -> Format-Custom
Alias             fhx -> Format-Hex

Version          Source
-----
3.1.0.0          Microsoft.PowerShell.Utility
3.1.0.0          Microsoft.PowerShell.Utility
```

Questions:

1. What is the alias set for the ipconfig.exe command?
2. Find the Execution Policy set for the LocalMachine scope.

Answer

Questions

Answer the question(s) below to complete this Section and earn cubes!

Target(s): 10.129.61.46 (ACADEMY-MISC-WS01) 🚀

Life Left: 55 minute(s) + Terminate ✕

🔗 RDP to 10.129.61.46 (ACADEMY-MISC-WS01) with user "htb-student" and password "Academy_WinFun"

+ 0 🟢

What is the alias set for the ipconfig.exe command?

ipconfig

Submit

+ 0 🟢

Find the Execution Policy set for the LocalMachine scope.

Unrestricted

Cheat Sheet

Download VPN Connection File

✓ Success

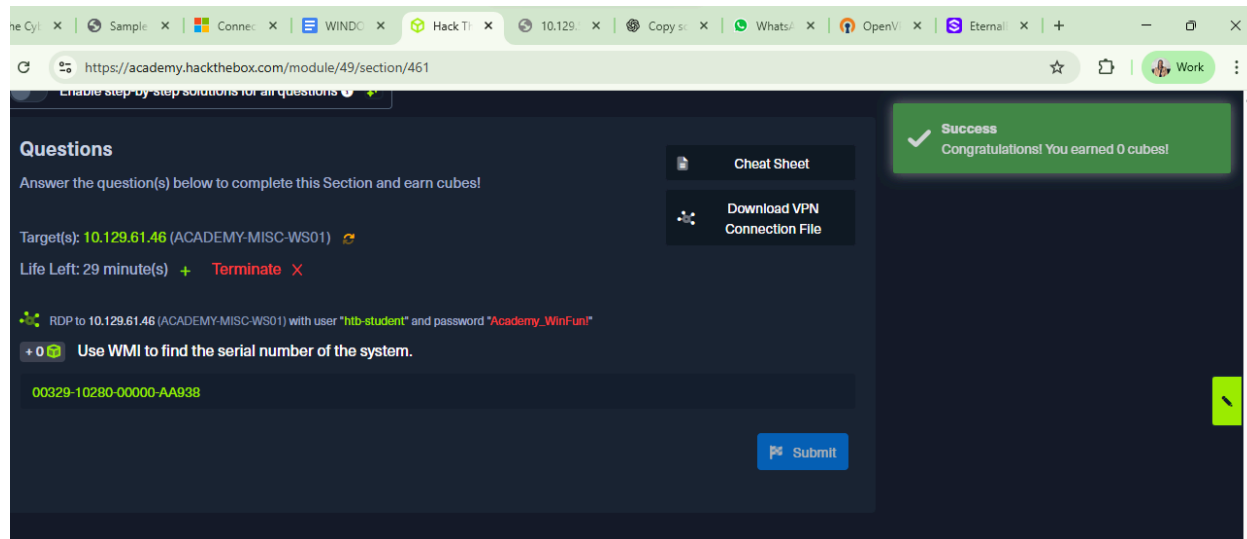
Congratulations! You earned 0 cubes!

1.8 Windows Management Instrument(WMI)

Windows Management Instrumentation (WMI) is a Microsoft framework that allows administrators and scripts to access, query, and manage information about Windows-based systems, including hardware, software, and operating system components.

Question: Use WMI to find the serial number of the system?

Answer:



1.9 Microsoft Management Console

Microsoft Management Console (MMC) is a Windows tool that provides a centralized interface to manage and configure different system settings and administrative tasks. It works like a “container” where you can add small management tools called snap-ins—for example, Event Viewer, Device Manager, or Services. Using MMC, administrators can view logs, manage users, monitor hardware, and configure services all in one window, instead of opening multiple separate tools. It’s especially useful for system administration and troubleshooting because it organizes different management functions in a consistent, easy-to-use interface.

1.10 Windows Subsystem for Linux (WSL)

Is a feature in Windows that lets you run a **Linux environment directly on Windows** without using a virtual machine or dual boot. It allows you to install and run Linux distributions like Ubuntu or Debian, use Linux command-line tools, scripts, and applications, and even access Linux files from Windows. WSL is mainly used by developers, sysadmins, and security professionals to combine the power of Linux tools with the Windows operating system, making it easy to work across both environments seamlessly.

1.11 Desktop Experience vs. Server Core are two installation options for Windows Server, designed for different use cases.

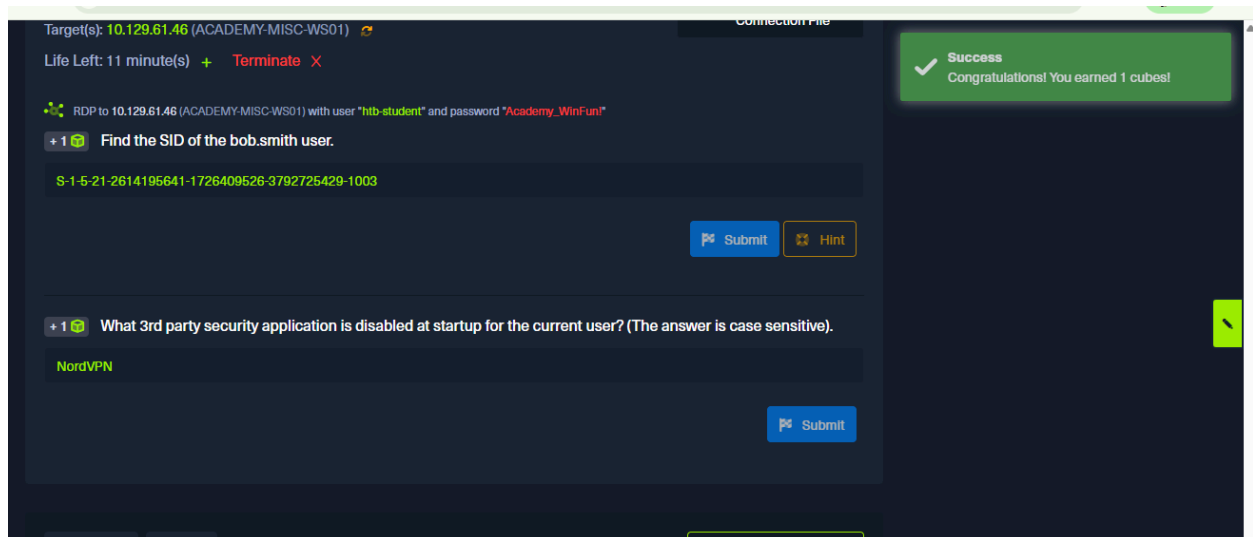
- **Desktop Experience** installs Windows Server with the **full graphical user interface (GUI)**, including File Explorer, Control Panel, and other desktop features. This makes it easier to manage for administrators who prefer a visual interface, but it uses more system resources and increases the attack surface.
- **Server Core** installs Windows Server **without the GUI**, providing only a minimal command-line interface (PowerShell or CMD). It consumes fewer resources, has a smaller footprint, and is more secure because there are fewer components to patch or exploit. Server Core is preferred in production environments where servers are managed remotely or automated via scripts.

1.12 Windows Security

Is the built-in security suite in Windows that helps protect your computer from viruses, malware, and other threats. It includes features like real-time protection, firewall and network protection, account protection, device security, and family options, all in one interface.

Question:

1. Find the SID of the bob.smith user.
2. What 3rd party security application is disabled at startup for the current user?

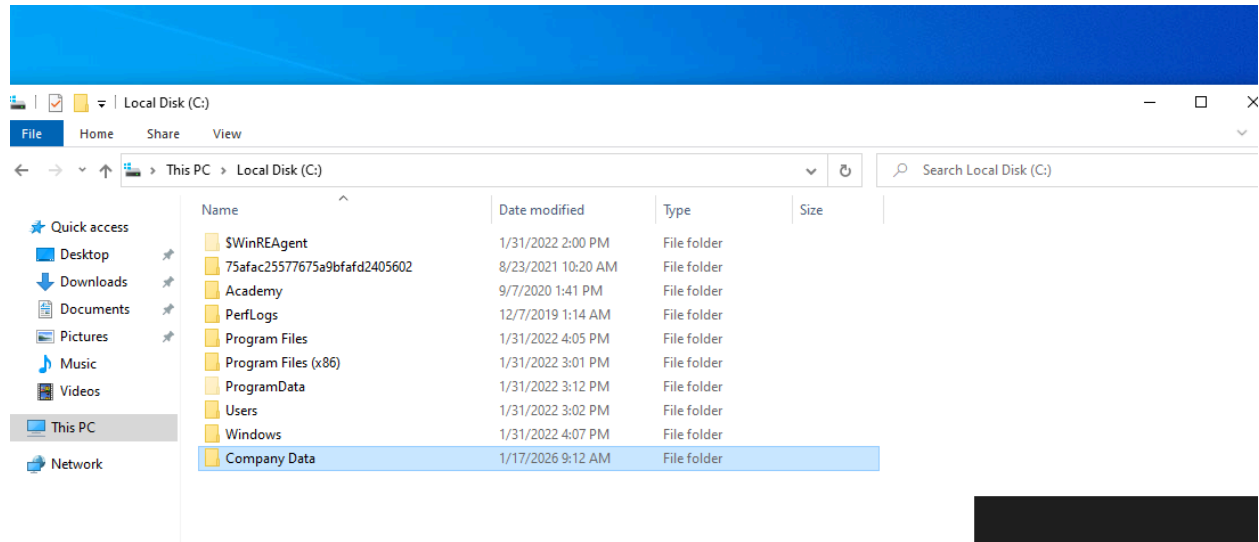


1.13 Skills Assessment -Window Fundamental

In this section , i was tasked on creating folders and sub-folders, giving permissions, creating groups, users and removing permissions

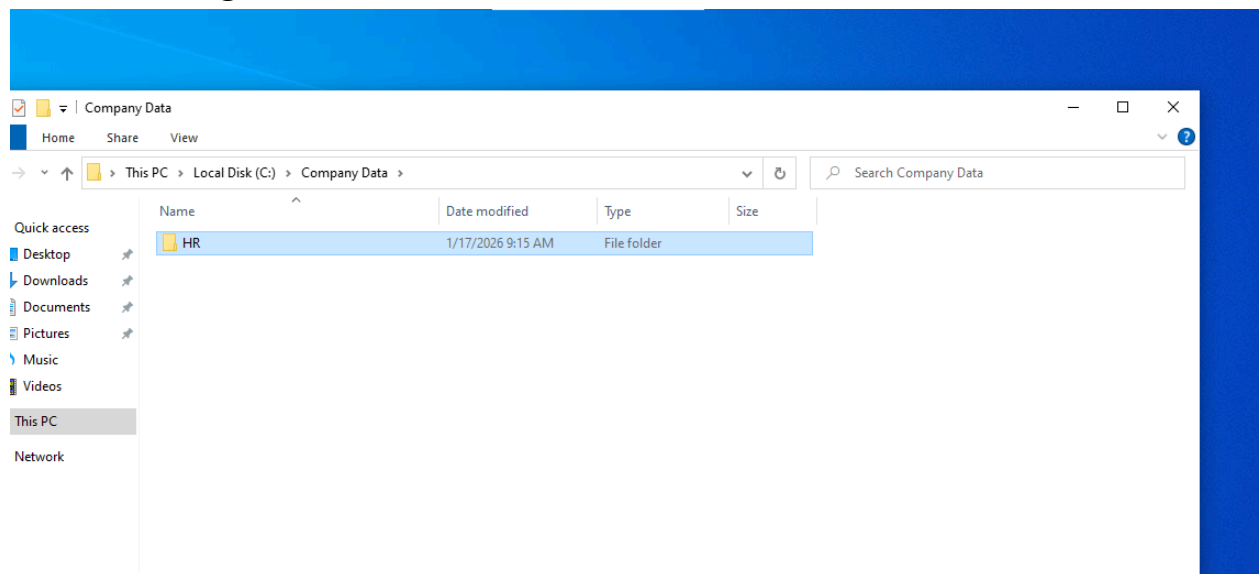
a)Creating a shared folder called Company Data

Below is the diagram to illustrate:



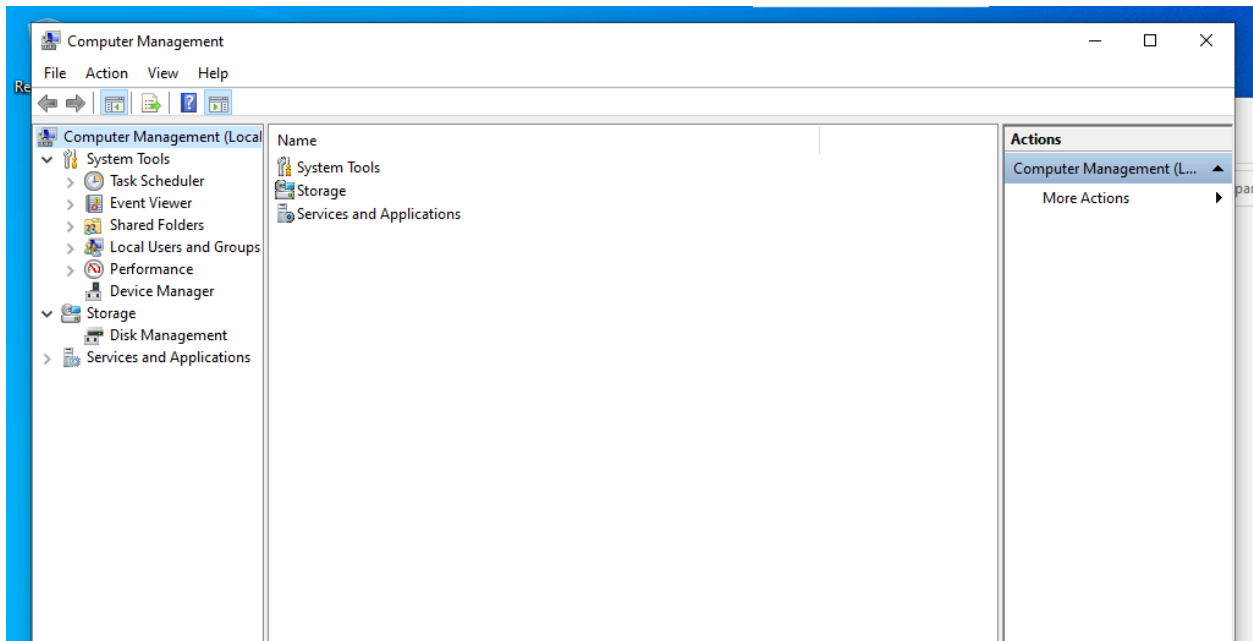
b)Creating HR inside the company data

Below is the diagram to illustrate:

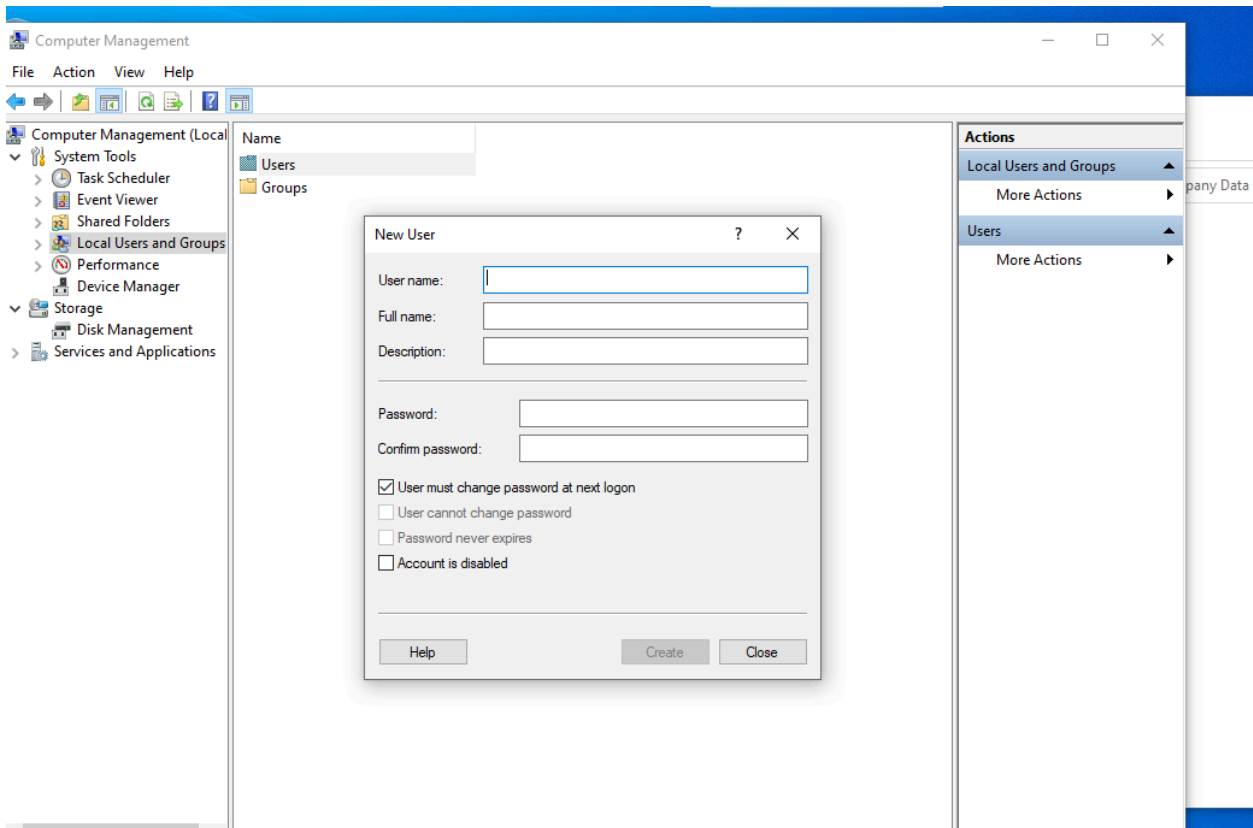


c) Create a new user Jim

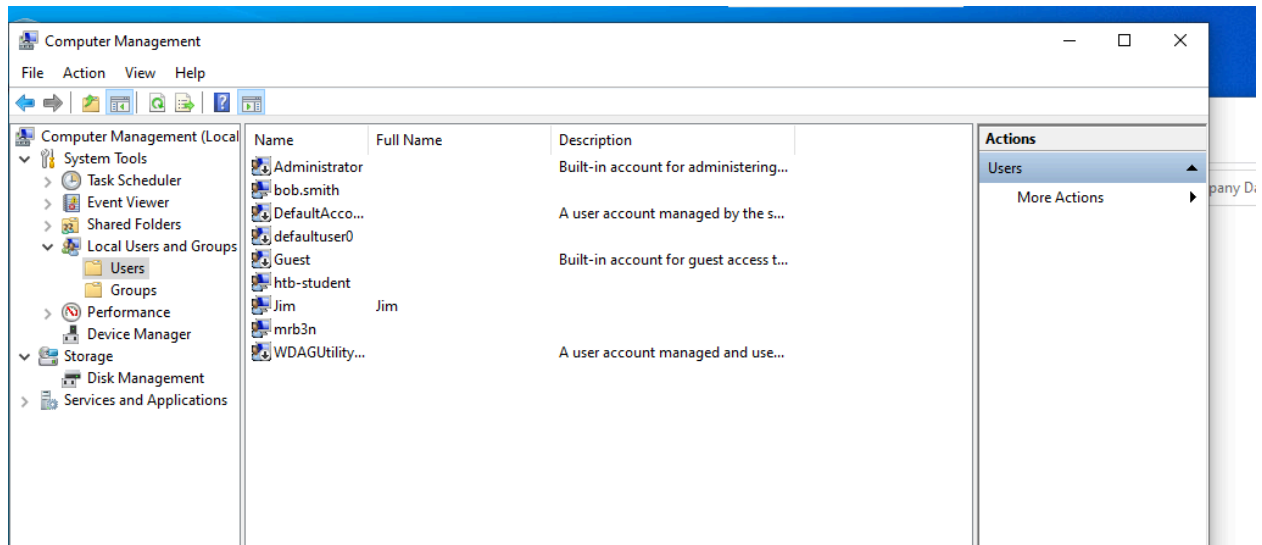
In this case I will go to my workstation, and open computer management. Then go to local users and groups then create new users. Below is the diagram to illustrate:



I clicked on users for this to appear where I am going to input Jim. Below is the diagram to illustrate:

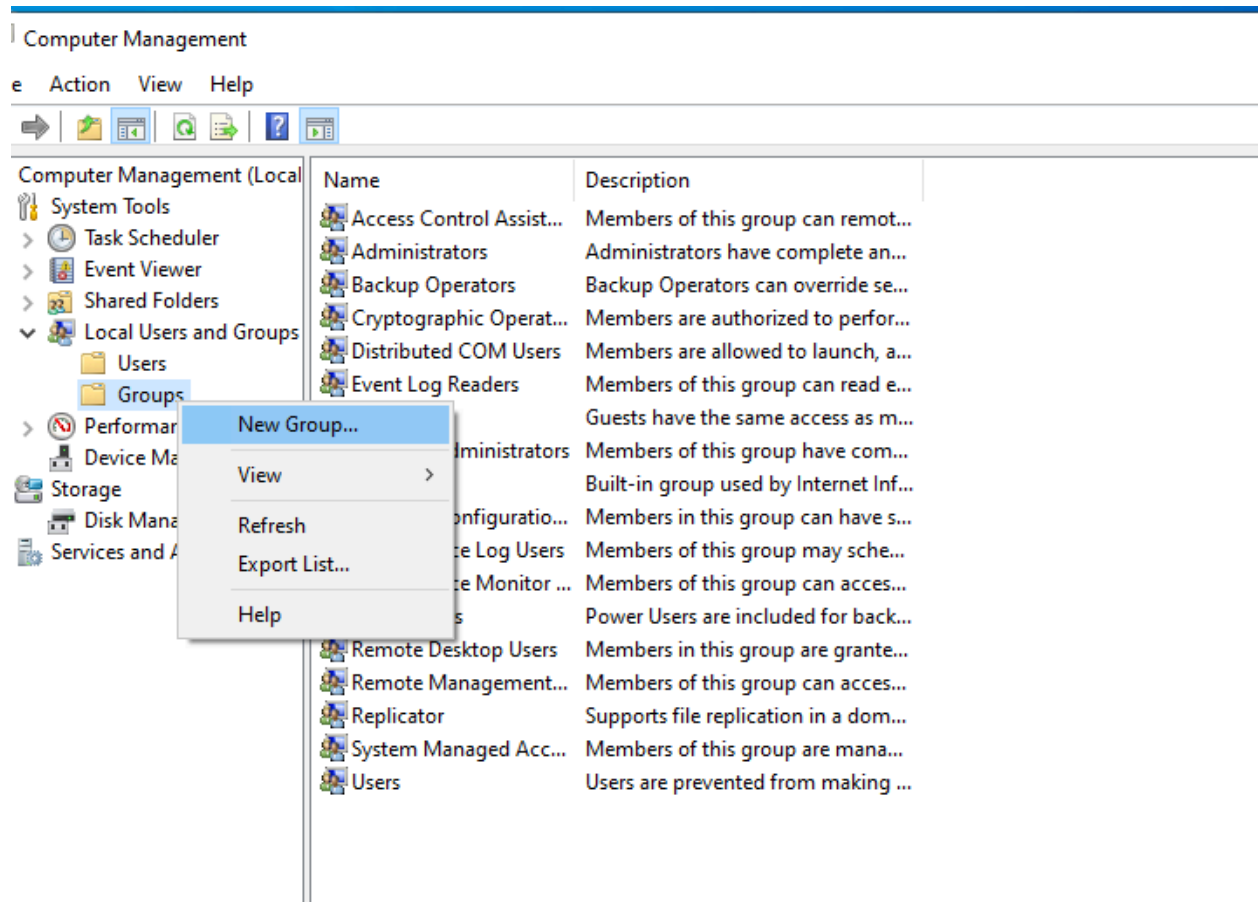


This is evidence that jim has been created. Below is the diagram to illustrate:

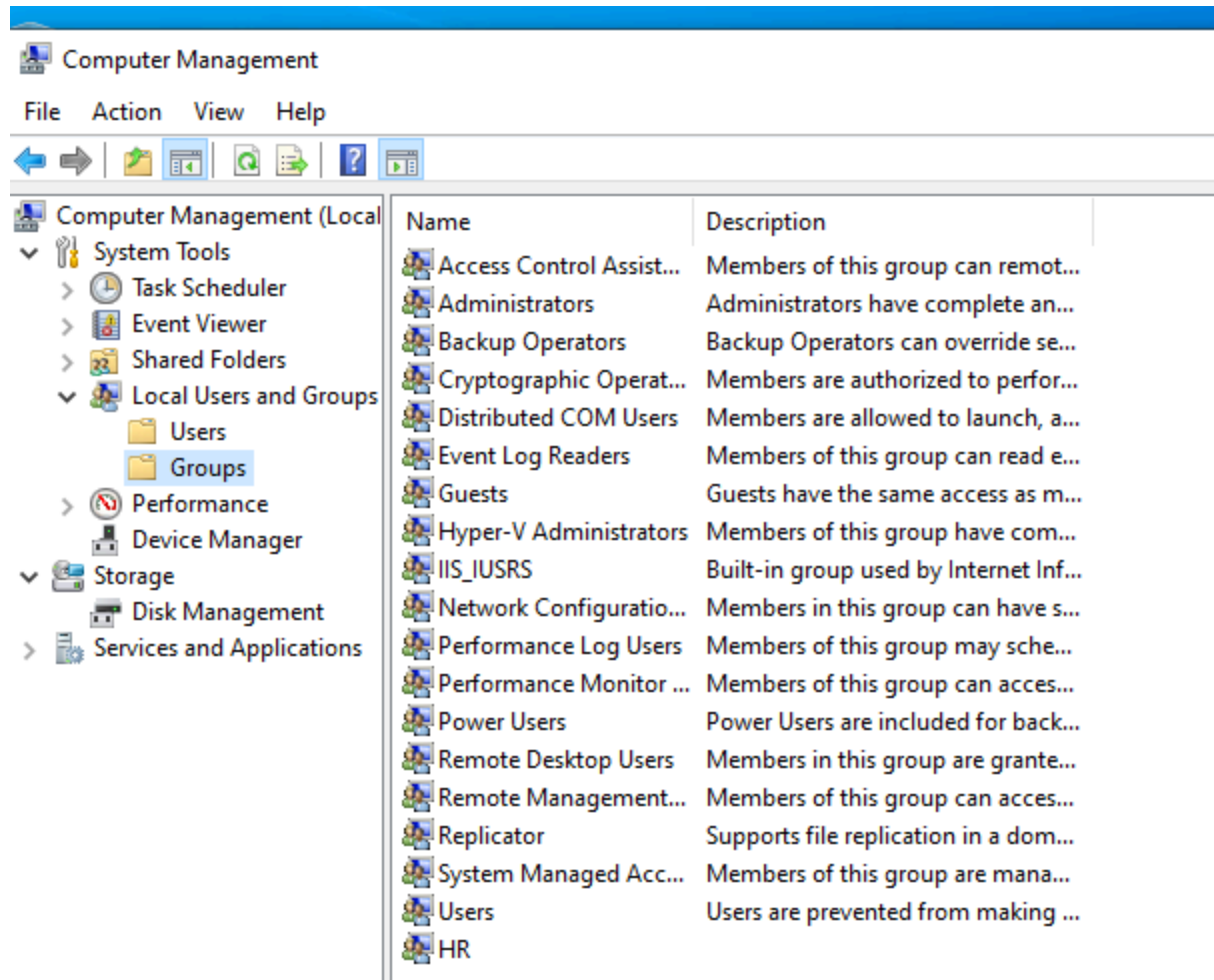


d) Creating a security group called HR

From my workstation, i will double-click on groups and create new, as shown below:

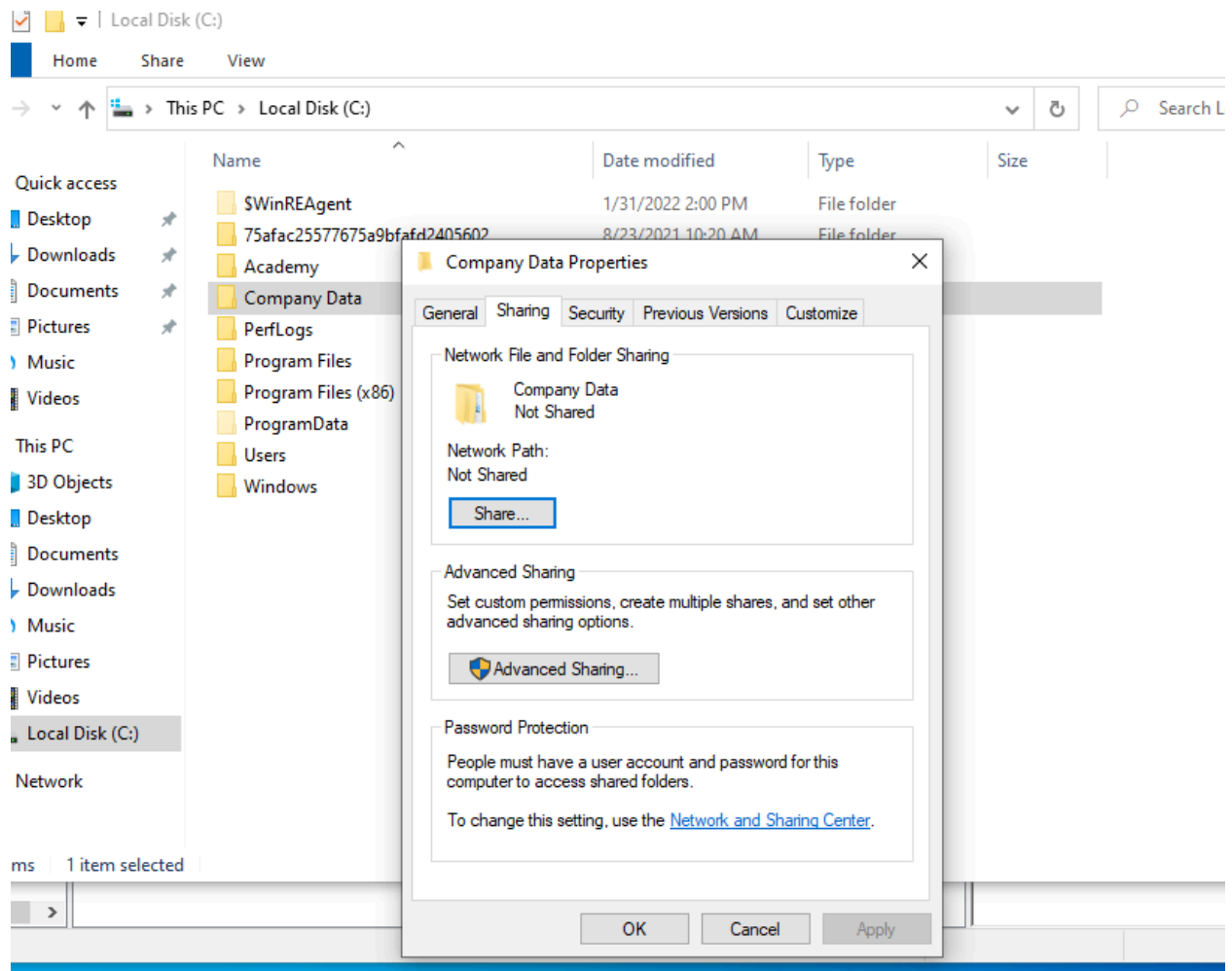


Then I will create the HR group. Below is the diagram to illustrate:

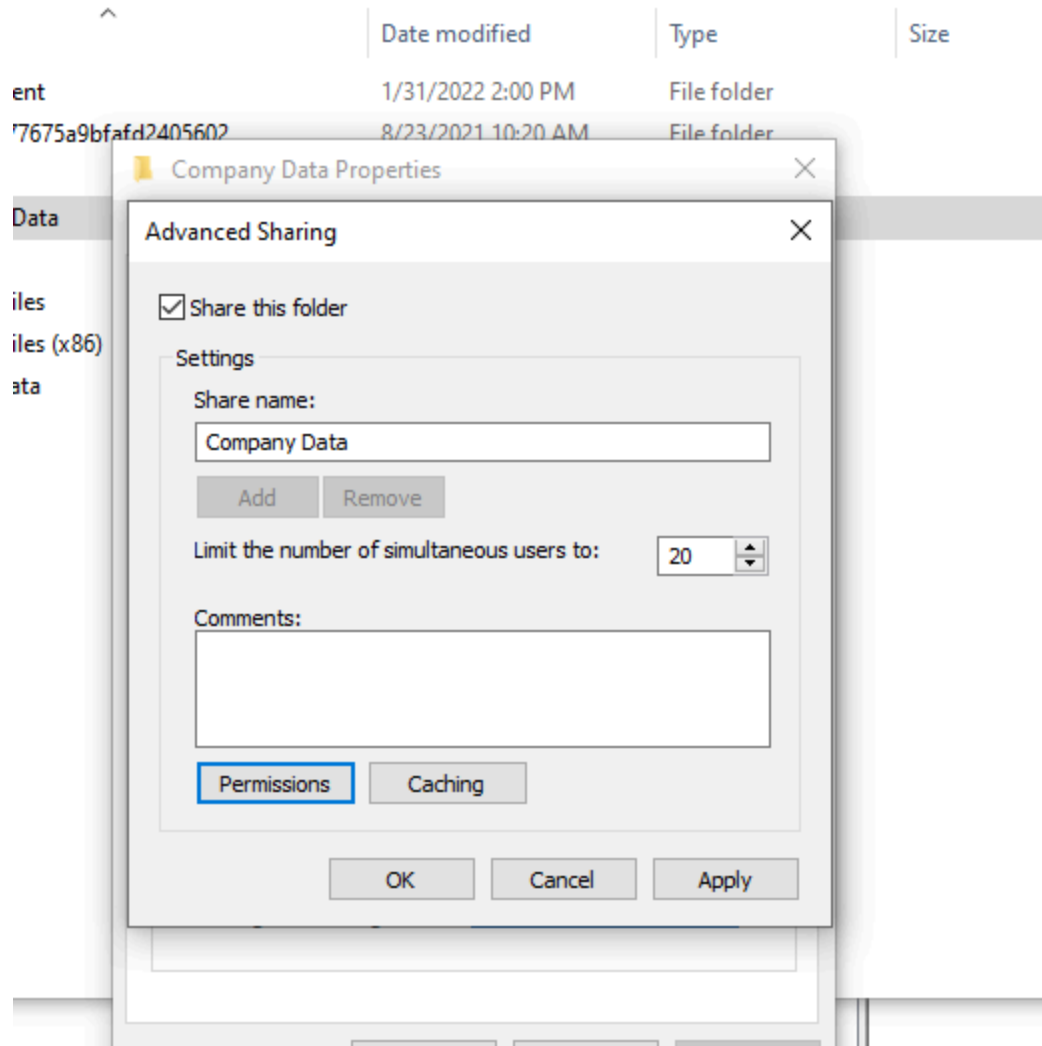


e)Adding the HR security group to the shared Company Data folder and NTFS permissions list.

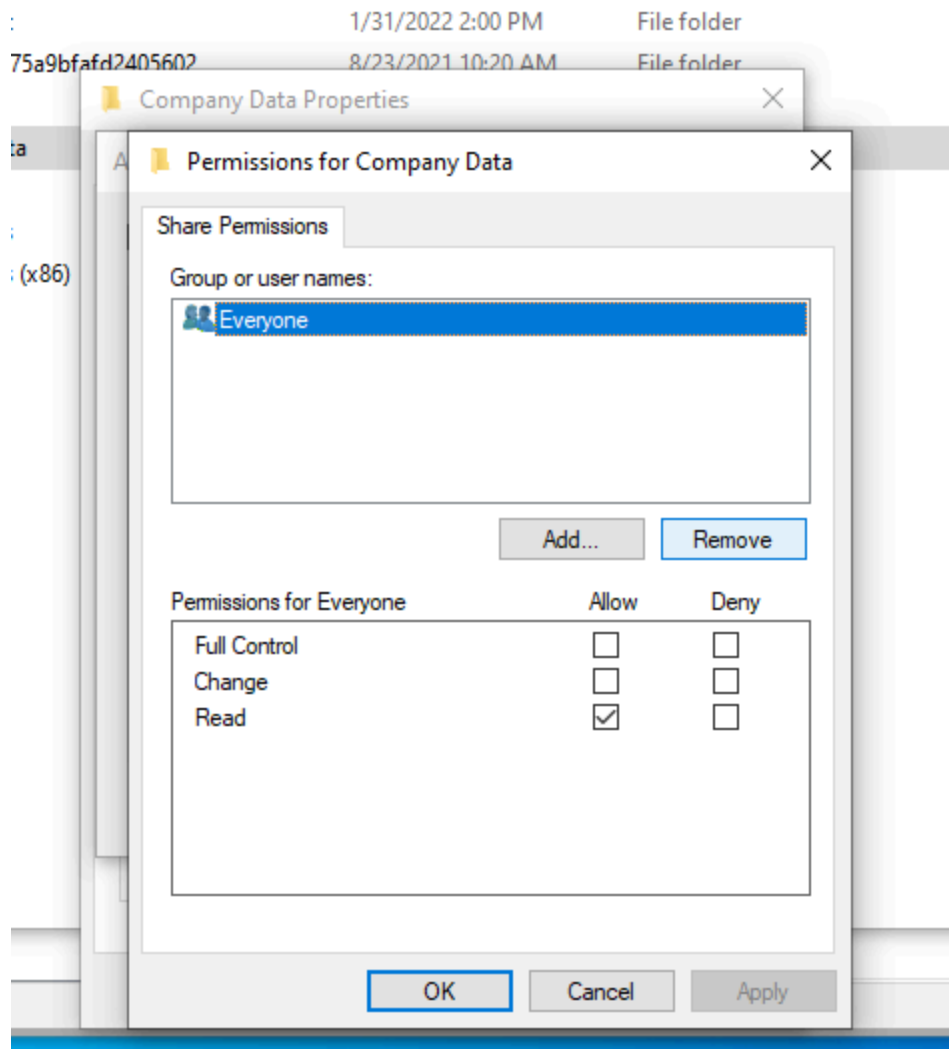
I will double click the Company Data folder, then go to properties, under properties i will click sharing.Below is the diagram to illustrate:



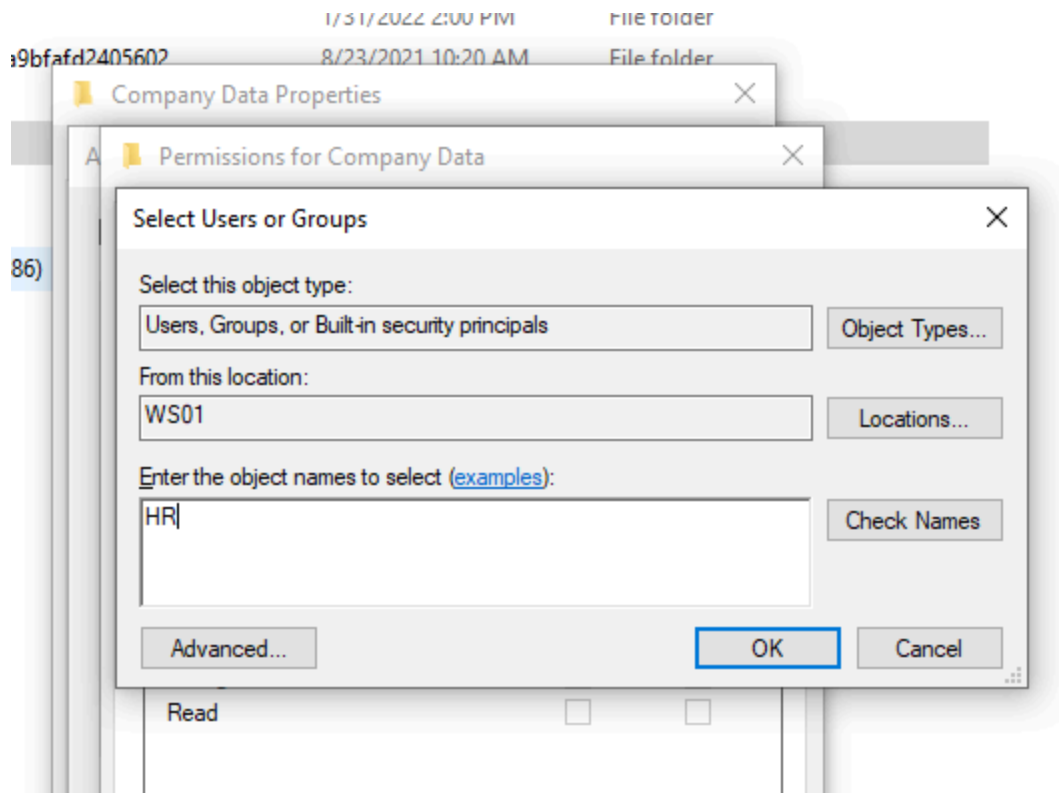
I will click advance sharing, i click on the share this file , then permissions.Below is the diagram to illustrate:



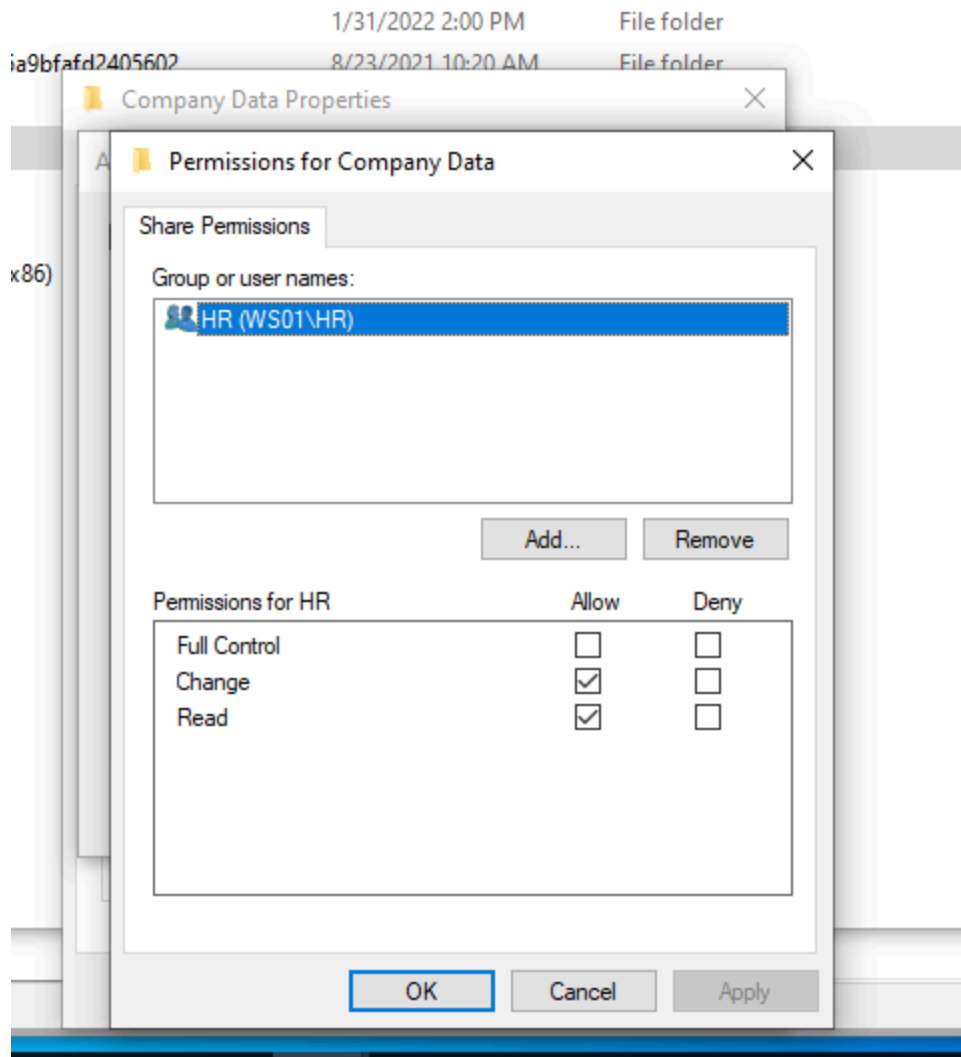
Then remove permissions for everyone. Below is the diagram to illustrate:



Then add HR to the group. Below is the diagram to illustrate:

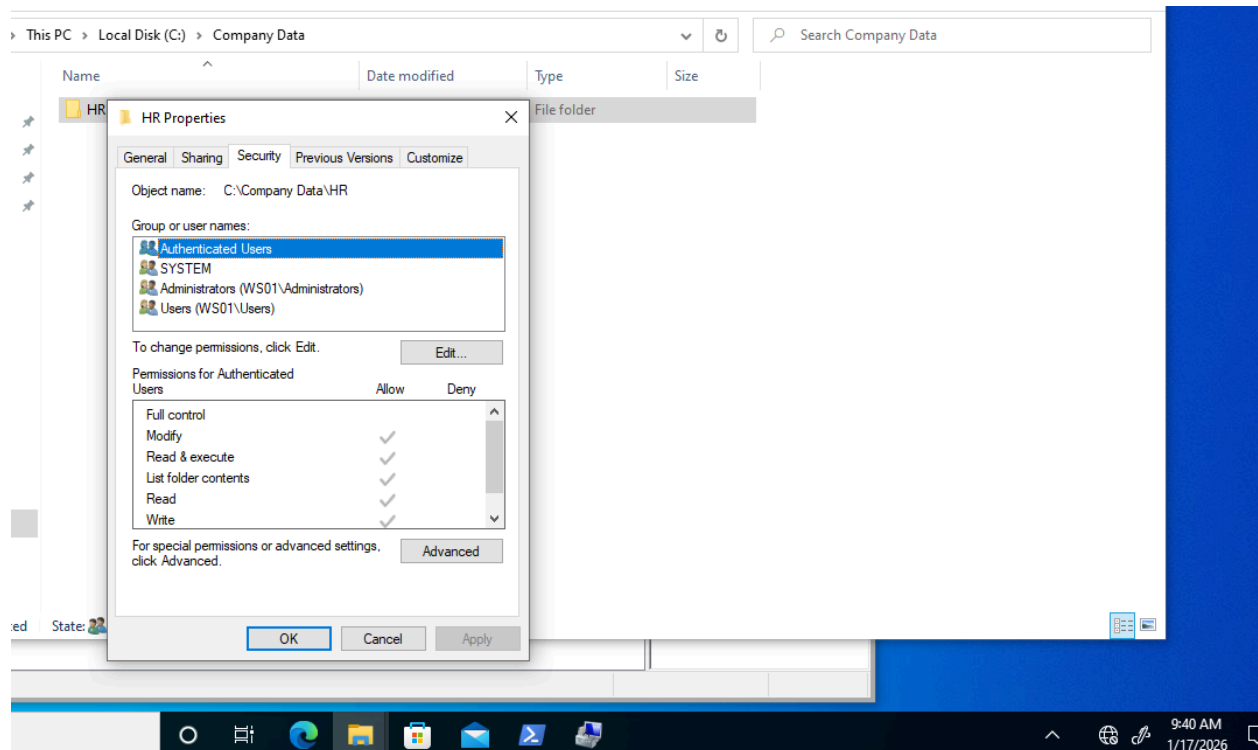


Then give HR permissions , to modify, read and write as shown:

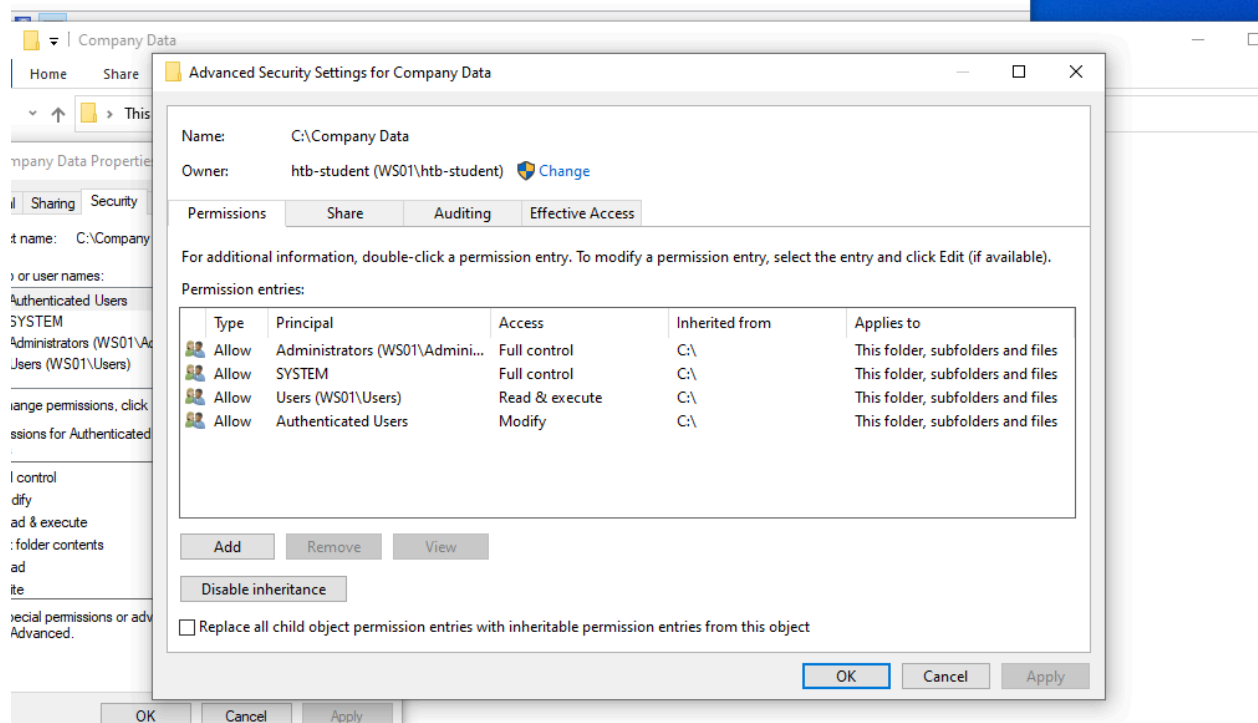


f) Adding the HR security group to the NTFS permissions list of the HR subfolders
Right-click the HR subfolder

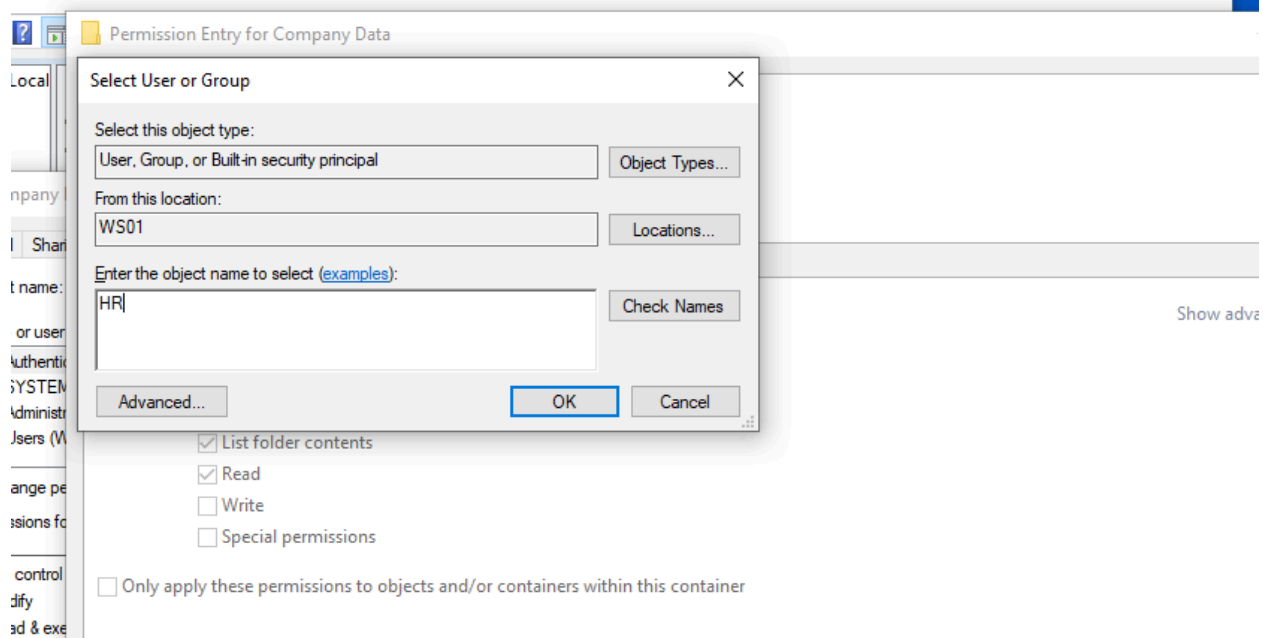
Select Properties → Security → Advanced



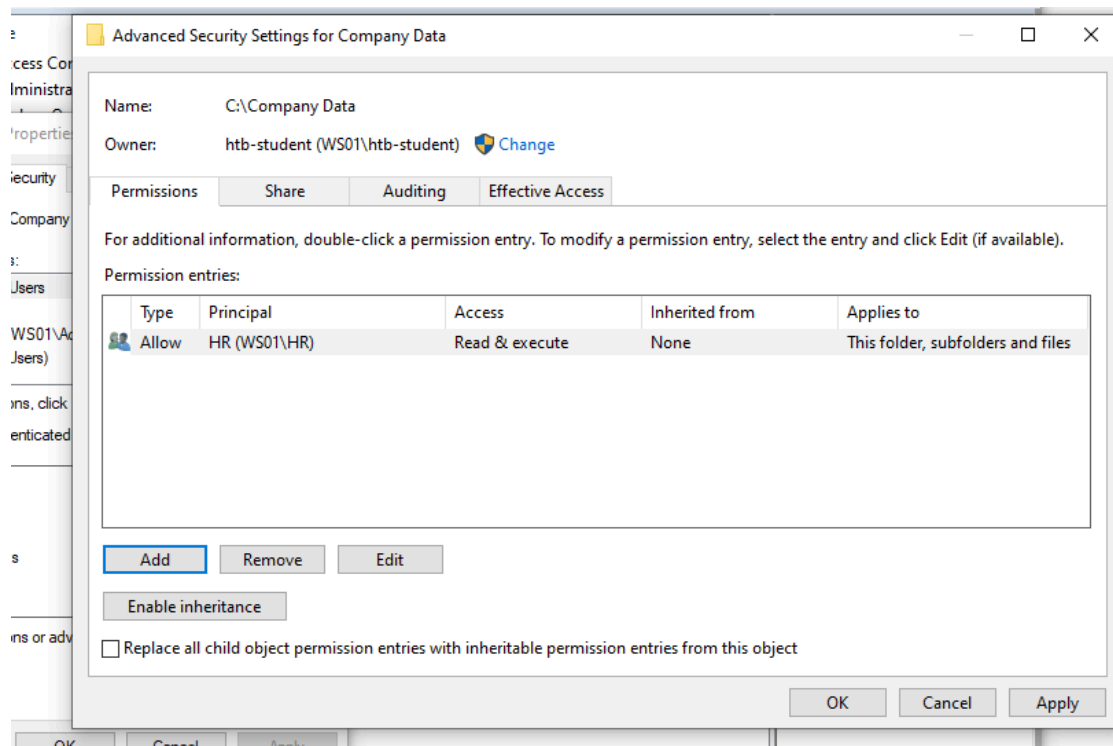
Then click on the advance button and disable inheritance.Below is the diagram to illustrate:



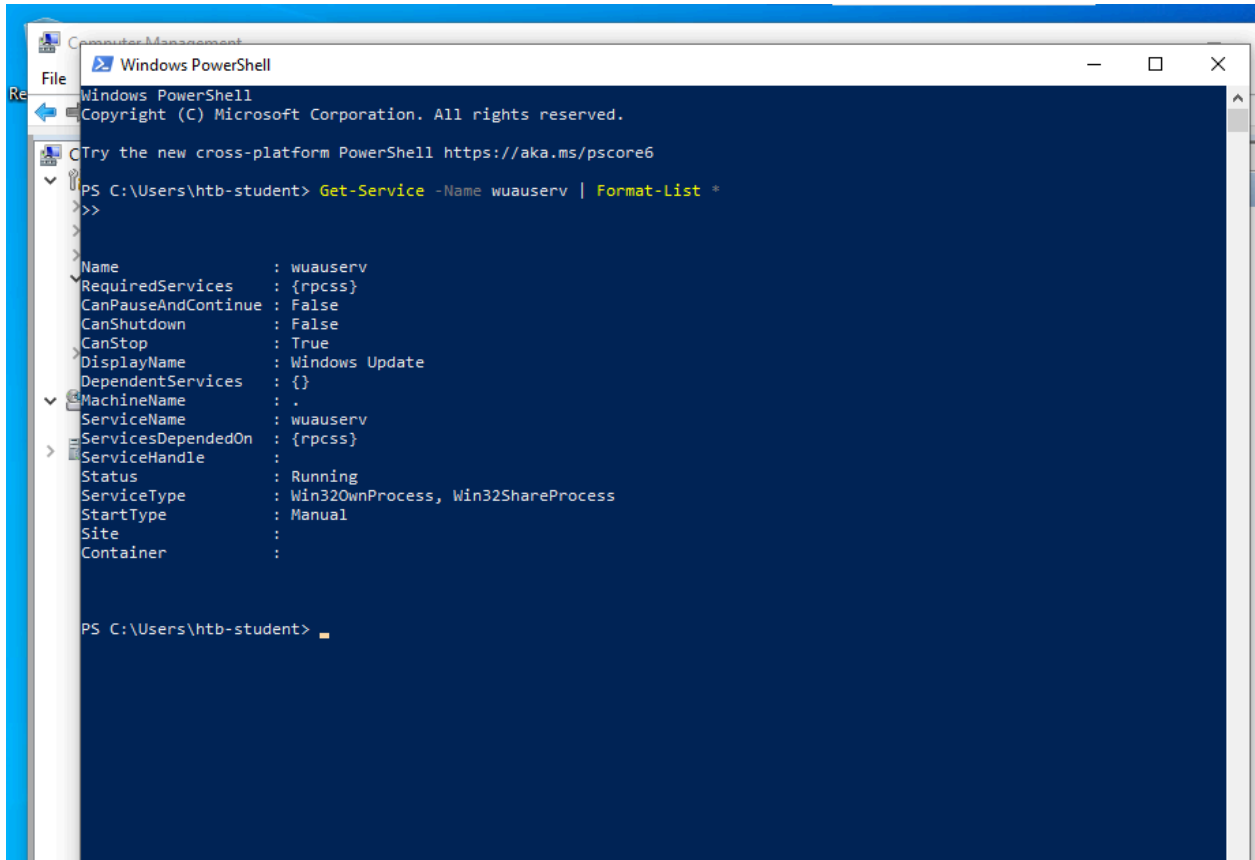
After it has enable,Add HR.Below is the diagram to illustrate:



HR has been added, Below is the diagram to illustrate:



g)Using Powershell to list details about a service.



```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Try the new cross-platform PowerShell https://aka.ms/pscore6

PS C:\Users\htb-student> Get-Service -Name wuauserv | Format-List *
Name                : wuauserv
RequiredServices     : {rpcss}
CanPauseAndContinue  : False
CanShutdown          : False
CanStop              : True
DisplayName           : Windows Update
DependentServices    : {}
MachineName          : .
ServiceName          : wuauserv
ServicesDependedOn   : {rpcss}
ServiceHandle        :
Status               : Running
ServiceType           : Win32OwnProcess, Win32ShareProcess
StartType             : Manual
Site                 :
Container            :
```

Questions:

1. What is the name of the group that is present in the Company Data Share Permissions ACL by default?

Answer: Everyone

2. What is the name of the tab that allows you to configure NTFS permissions?

Answer: Security

3. What is the name of the service associated with Windows Update?

Answer: wuauserv

4. List the SID associated with the user account Jim you created.

Answer: S-1-5-21-2614195641-1726409526-3792725429-1006

5. List the SID associated with the HR security group you created.

Answer: S-1-5-21-2614195641-1726409526-3792725429-1007

Answers ,Below is the diagram to illustrate:

Target(s): 10.129.61.0 (ACADEMY-MISC-WS01) 

Life Left: 106 minute(s) + **Terminate** X

RDP to 10.129.61.0 (ACADEMY-MISC-WS01) with user "htb-student" and password "Academy_WinFun!"

+ 1  What is the name of the group that is present in the Company Data Share Permissions ACL by default?

Everyone

Submit **Hint**

+ 1  What is the name of the tab that allows you to configure NTFS permissions?

security

Submit **Hint**

+ 1  What is the name of the service associated with Windows Update?

wuauserv

The Cyl X Sample X Connec X WINDC X 10.129. X Hack Ti X Copy s X Whats X OpenV X Eternali X +

→   https://academy.hackthebox.com/module/49/section/1015  

+ 1  List the SID associated with the user account Jim you created.

S-1-5-21-2614195641-1726409526-3792725429-1006

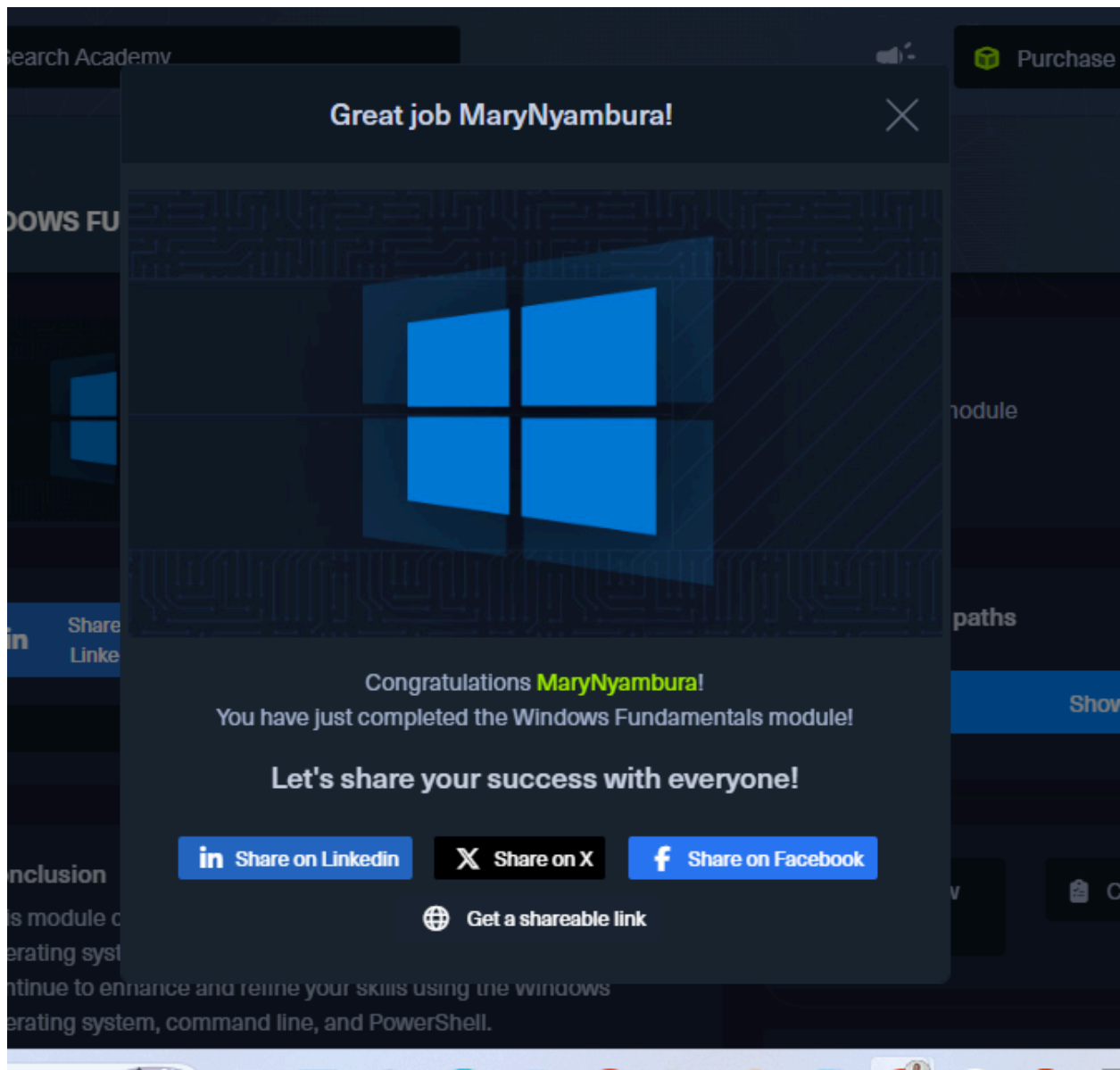
Submit **Hint**

+ 1  List the SID associated with the HR security group you created.

S-1-5-21-2614195641-1726409526-3792725429-1007

Submit **Hint**

I was able to finish the module and got awarded some cubes. Below is my congratulations screenshot:



<https://academy.hackthebox.com/achievement/2382167/49v>

Conclusion and summary

I gained a solid understanding of the Windows operating system, including the file system, directory traversal, and permission management. I learned how to manage users and groups, assign and control access rights, and work with system tools such as Task Manager, Computer Management, and Event Viewer. This knowledge is essential for a security analyst, as it builds the foundation for securing Windows environments, identifying misconfigurations, monitoring services, and detecting unauthorized access using tools like PowerShell, WMI, and system auditing features.